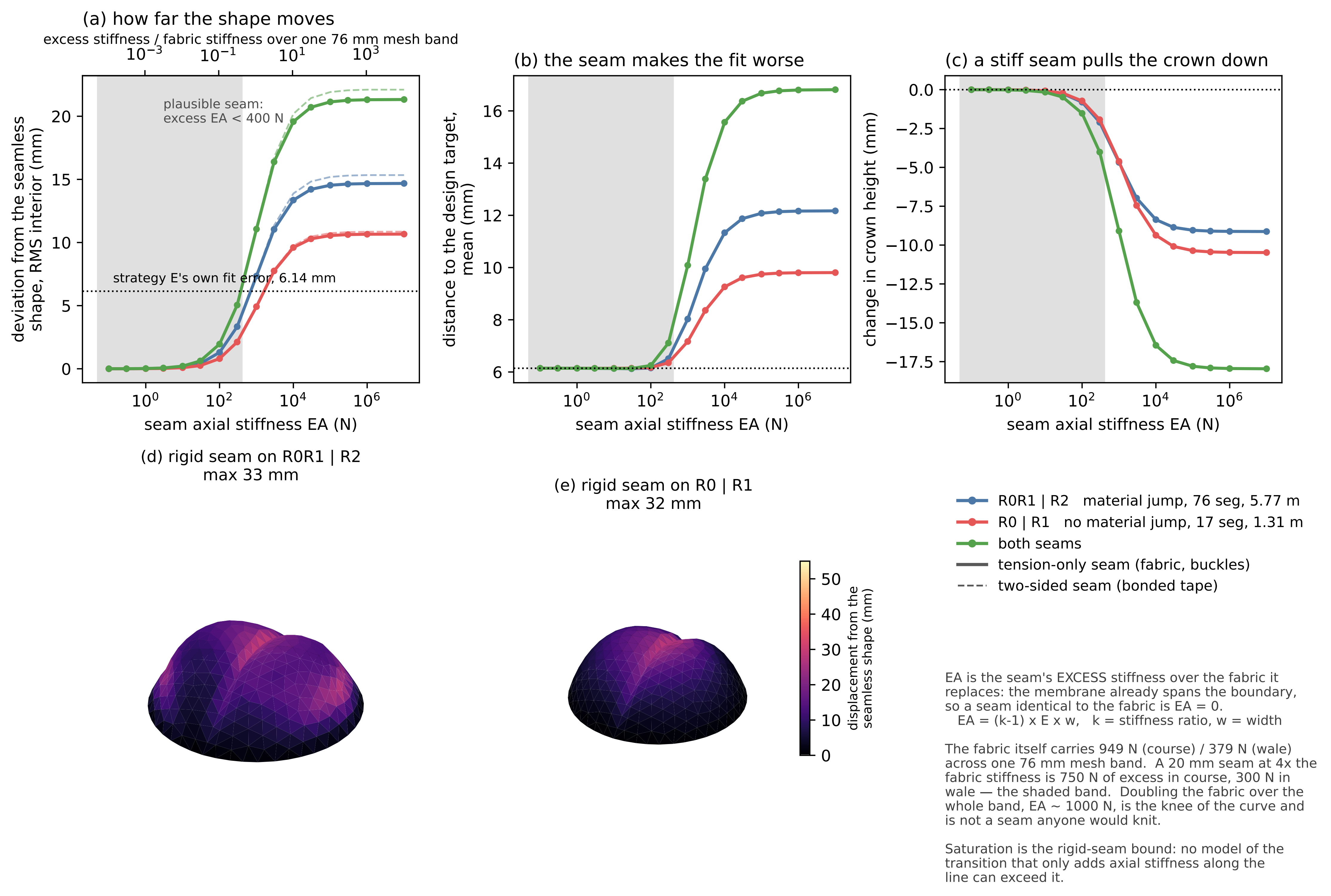
(a) how far the shape moves
excess stiffness / fabric stiffness over one 76 mm mesh band
10^{-3} 10^{-1} 10^{1} 10^{3}
plausible seam:
excess EA < 400 N
strategy E's own fit error, 6.14 mm
deviation from the seamless shape, RMS interior (mm)
seam axial stiffness EA (N)
(b) the seam makes the fit worse
distance to the design target, mean (mm)
seam axial stiffness EA (N)
(c) a stiff seam pulls the crown down
change in crown height (mm)
seam axial stiffness EA (N)
(d) rigid seam on R0R1 | R2
max 33 mm
(e) rigid seam on R0 | R1
max 32 mm
displacement from the seamless shape (mm)
R0R1 | R2 material jump, 76 seg, 5.77 m
R0 | R1 no material jump, 17 seg, 1.31 m
both seams
tension-only seam (fabric, buckles)
two-sided seam (bonded tape)
EA is the seam's EXCESS stiffness over the fabric it replaces: the membrane already spans the boundary, so a seam identical to the fabric is EA = 0.
EA = (k-1) x E x w, k = stiffness ratio, w = width
The fabric itself carries 949 N (course) / 379 N (wale) across one 76 mm mesh band. A 20 mm seam at 4x the fabric stiffness is 750 N of excess in course, 300 N in wale — the shaded band. Doubling the fabric over the whole band, EA ~ 1000 N, is the knee of the curve and is not a seam anyone would knit.
Saturation is the rigid-seam bound: no model of the transition that only adds axial stiffness along the line can exceed it.